

Introduction to Data-Based Questions

Candidates will be required to process, visualise and analyse data using spreadsheet software. The use of spreadsheet software, including the ability to input correct functions with appropriate syntax, is one of the skills being assessed.

Each candidate will have access to a computer installed with spreadsheet software for the duration of the examination and a quick reference guide for the duration of the examination.

Similar to the use of calculators, there will be no assistance rendered to candidates during the exam for the use of spreadsheet software.

Requirements On Use Of Spreadsheet For Practical Assessment

Candidates should be able to:

- (a) import data files (in csv format) into the spreadsheet for subsequent processing
- (b) input data obtained from an experiment for subsequent processing

Functions Within The Spreadsheet

- (a) Perform mathematical operations as stipulated in the Mathematical Requirements, including sines, cosines, tangents (and the inverse functions) on angles in radians (converting from degrees as necessary), exponentials and logarithms (lg and ln), and use of scientific notation (e.g. 3.00E+14 for 3.00×10^{14} and 3E-4 for 0.0003).
- (b) Use features for numerical data manipulation
 - (i) use and input formulae and duplicate formulae across cells
 - (ii) adjust formatting (e.g. data type) and duplicate formatting across cells
- (c) Plot a labelled line graph using selected data from a table
 - (i) adjust the scale of both axes
 - (ii) use selected data points to give a line of best fit (using built-in functions to add a linear trendline)
 - (iii) display the equation of a trendline
 - (iv) determine the area under a curve (e.g. summation of rectangular or trapezoidal areas)
 - (v) determine the gradient at a point on a curve (e.g. calculating in a small interval near the point) $\Delta y / \Delta x$

Refer to SEAB website for the [H2 Physics 9478 Syllabus](#) and [Excel Reference Guide](#) for more information.

Steps for Analysing Large Datasets

1 Manage anomalous data

- (a)** Insert a scatter chart of y vs x to visually check for anomalous data.
- (b)** Hover the cursor over an anomalous data to determine its (x,y) coordinates.
- (c)** Anomalous data must be deleted from the table before further processing can take place.
Right-click on the row number that contains the anomalous data and select “Delete”.
Or, select the relevant cells and right-click to select “Delete...” and “Shift Cells Up”.
The anomalous point will disappear from the chart.
Do not leave empty rows within the table as this will produce erroneous results later.

2 Manage noise and systematic error

- (a)** Besides anomalies, large datasets may contain noise and systematic error.
- (b)** If noise is present in a supposedly constant set of data (e.g. a steady current), the constant value is found by using the average value of that set of data (after removing any anomaly).
- (c)** If a systematic shift in a column of data is detected, a corrected column of data can be calculated by adding or subtracting the systematic error. (E.g. background radiation.)

3 Linearise the given equation and compute the necessary dependent and independent variables.

4 Plot a straight-line graph

- (a)** Highlight the two columns of data to be plotted, including the titles (a whole column can also be selected by clicking on the column heading).
This can be done by pressing and holding the CTRL key, while using the mouse to select the cells or columns.
- (b)** The left column is always plotted along the horizontal axis and the right column along the vertical axis.
- (c)** Insert a scatter chart (not line chart).

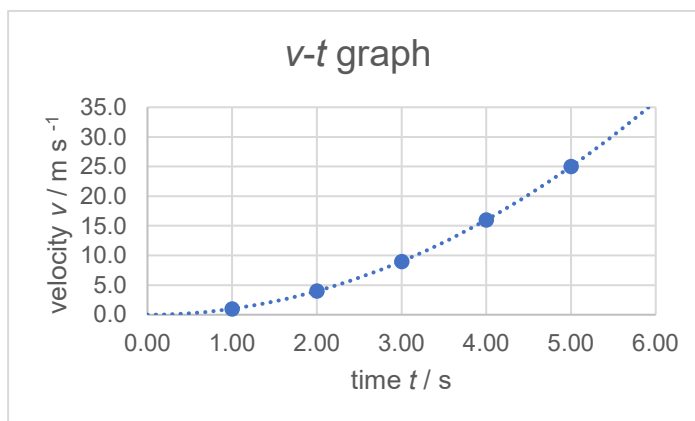
5 Insert the trendline and its equation (R^2 value is optional)

- (a)** Right-click on any plotted point, select “Add Trendline...”.
- (b)** Then select “Linear”, “Display Equation on Chart” and “Display R-squared Value on Chart”.
An R^2 value that is close to 1 indicates that the data fits the model (optional).
You may shift the equation and increase its font size if you have difficulty viewing it.
- (c)** Set the numbers in the equation to at least 3 s.f. by adjusting the d.p. of the numbers.

The data for these practices is found in the “Introduction” tab in the digital file: datasetWS14.xlsx

Practice 1: Determine Gradient at a Point on a Curve

	A	B
1	time t / s	velocity v / m s^{-1}
2	0.00	0.0
3	1.00	1.0
4	2.00	4.0
5	3.00	9.0
6	4.00	16.0
7	5.00	25.0



[Note: You may use the cursor or the mouse to reference a cell instead of typing it.]

To determine acceleration from a v - t curve:

Gradient can be determined using “= SLOPE(known_ys, known_xs)”.

Gradient at 2.00 s can be found using this formula in cell C9: “=SLOPE(B3:B5,A3:A5)”.

Value of cell C9: 4.00 m s^{-2}

Gradient at 4.50 s can be found using this formula in cell C10: “=SLOPE(B6:B7,A6:A7)”.

Value of cell C10: 9.00 m s^{-2}

Practice 2: Determine Area under a Curve

	A	B	C	D
1	time t / s	velocity v / m s^{-1}	area under curve	
2	0.00	0.0		
3	1.00	1.0		
4	2.00	4.0	=0.5*(A4-A3)*(B4+B3)	← area of trapezium between 1-2 s
5	3.00	9.0	=0.5*(A5-A4)*(B5+B4)	← area of trapezium between 2-3 s
6	4.00	16.0	=SUM(C4:C5)	← total area of trapeziums between 1-3s
7	5.00	25.0		

[Note: You may use the cursor or the mouse to reference a cell instead of typing it.]

To determine displacement from a v - t curve:

Area under the curve between 1.00 s to 3.00 s can be found by adding the areas of two trapeziums in cell C6 as shown above.

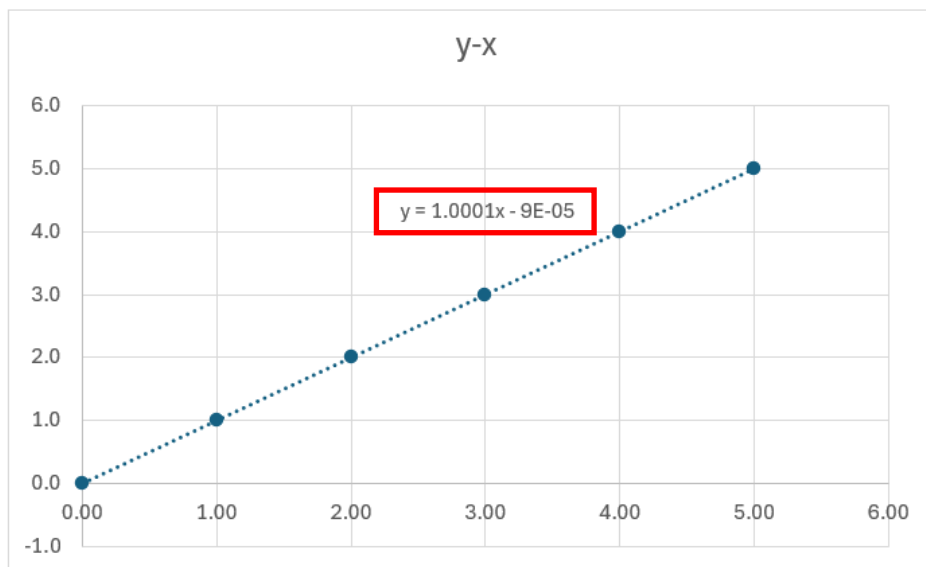
Value of cell C4: 2.50 m

Value of cell C5: 6.50 m

Value of cell C6: 9.00 m

Practice 3: Increase the Number of s.f. in the Equation of Trendline

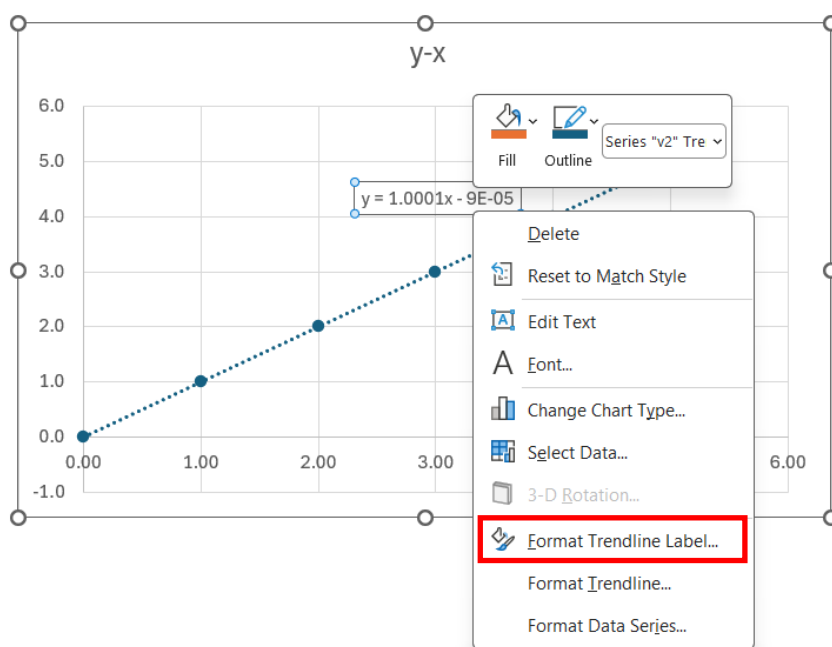
An "E" in Excel stands for exponent and indicates scientific notation, representing "times 10 to the power of". In the equation of the trendline below, the vertical intercept of "9E-05" represents 9×10^{-5} .



To increase the number of s.f. for the gradient or vertical intercept:

1. right-click on the equation
2. select **"Format Trendline Label ..."** to bring up dialogue box on the right-side of the screen
3. in **"Label Options"**, under **"Category"**, select either **"Number"** or **"Scientific"**
4. key in the desired number of decimal places

Step 1. & 2.



Step 3. & 4.

Write the equation of the trendline, with the gradient and the vertical-intercept to 3 s.f., when

"Number" is selected and set to 7 d.p.: $y = 1.00 x - 0.0000857$ (note the update in question)

"Scientific" is selected and set to 2 d.p.: $y = 1.00 x - 8.57 \times 10^{-5}$ (note the update in question)